SPIDERROCK
PLATFORM

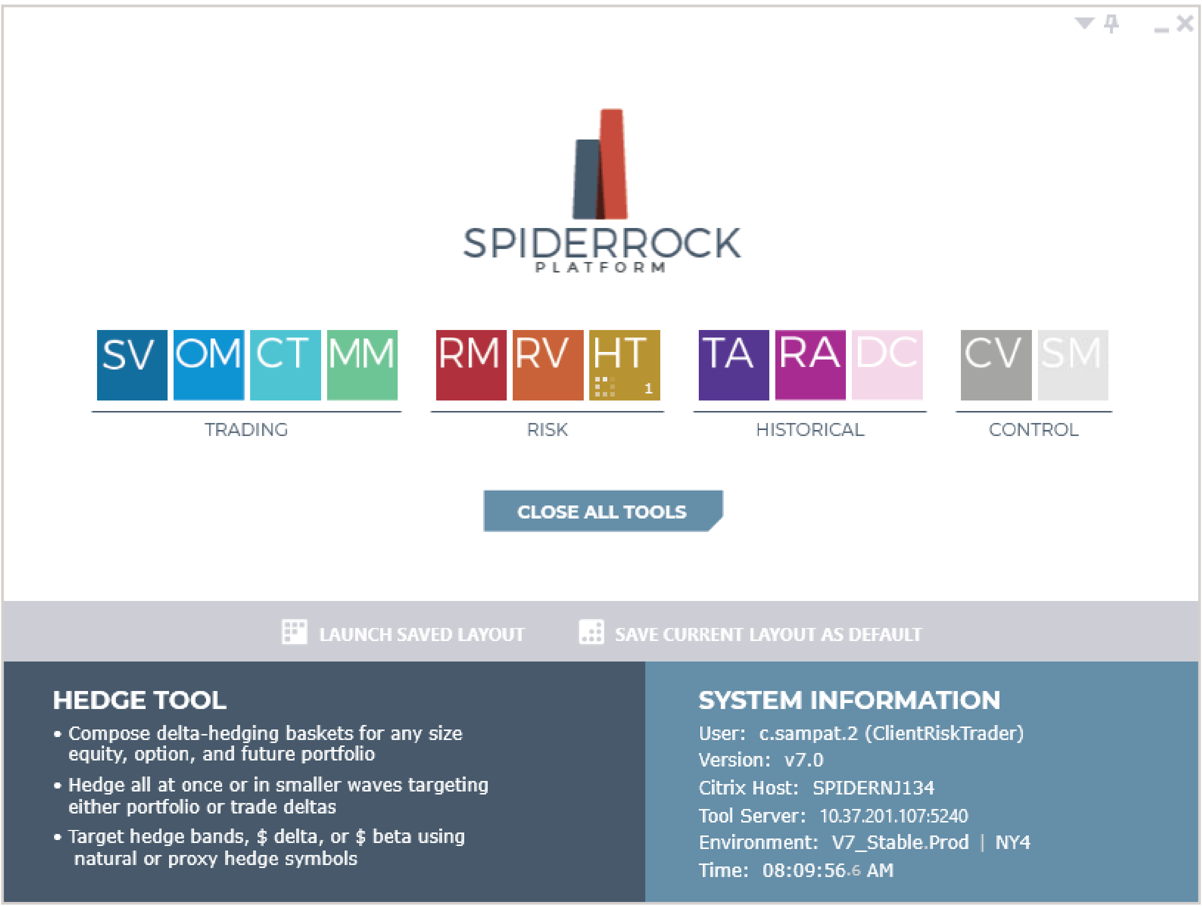
SpiderRock V7 Tool Concept Guide

Hedge Tool

March 16, 2021

OVERVIEW

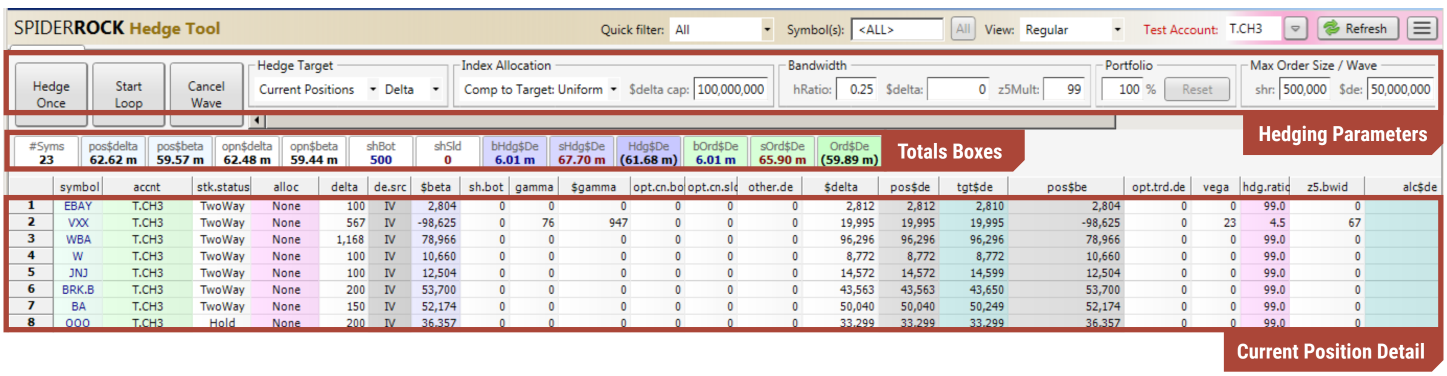
Versatile tool allowing customers to dynamically hedge any portfolio or sections of a portfolio. Users can hedge the delta or beta of their current positions, open positions or day trades using natural or proxy symbols. Users can also mitigate the cost of hedging by setting a risk tolerance limiting the hedging to the nearest delta band, and by choosing a mid market execution algo. Users can choose to hedge the portfolio all at once or send subsequent waves to automatically decrement the delta exposure to the desired level over a user defined period of time.



HEDGE TOOL FUNCTIONALITIES

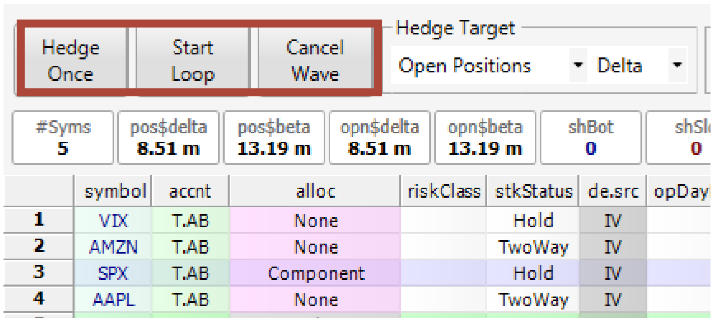
The Hedge Tool is comprised of three sections: Hedging Parameters, Totals Boxes, and Current Position Detail. Each section is summarized below and described in more detail on the following pages.

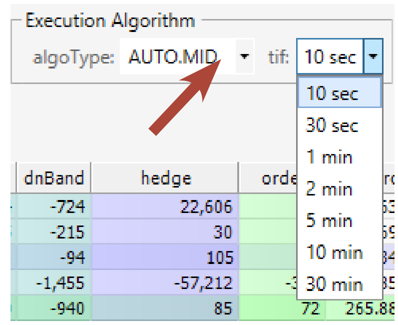
- Hedging Parameters allow users to set hedge target, index allocation, bandwidth (risk tolerance), portfolio size, max order size/wave and execution algorithms.
- Totals Boxes summarize the current position risk associated with the portfolio or the portion of portfolio displayed in the Current Position Detail window. It also indicates how much dollar delta is being bought or sold (green boxes).
- Current Position Detail displays risk metrics associated to start of day positions plus any day trades at the root symbol level.



HEDGING PARAMETERS

- Users can hedge the portfolio once or set waves that will work for a user defined timein-force. This method will decrement the deltas until the nearest risk band is met or the user cancels the wave.

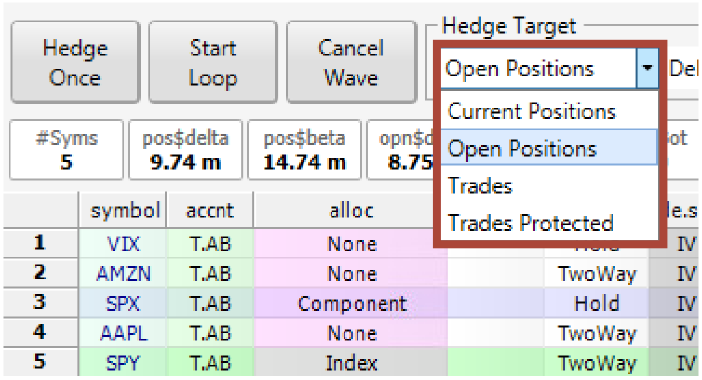


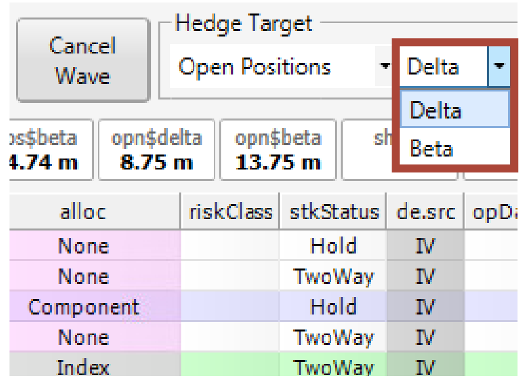


- **Hedge Once Button:** Users can choose to hedge once. The basket of orders will work for the time-in-force (tif) selected.

- **Start Loop:** Users can send waves of orders and each wave will work for the time-in-force selected. This would decrement the deltas in a looping fashion until manually canceled or until hedging is complete.

- **Cancel Wave:** This will prompt the system to stop any waves of orders that were previously triggered by the user via the Start Loop button.





- **Hedge Target**

  Users can set the hedge target to the following options:

  - Current Positions: Hedging Start of Day (SOD) positions plus any day trades
  - Open Positions: Hedging Start of Day (SOD) positions only
  - Trades: Hedging day trades only. SOD positions are not taken into account
  - Trades Protected: Hedging day trades only, but will not hedge trades that offset current position risk

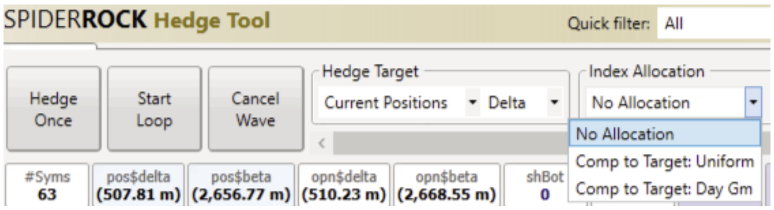
  Users have a choice between the following options for hedging based on delta or beta:

  - Delta: The hedging target is based on $ deltas
  - Beta: The hedging target is based on $ betas

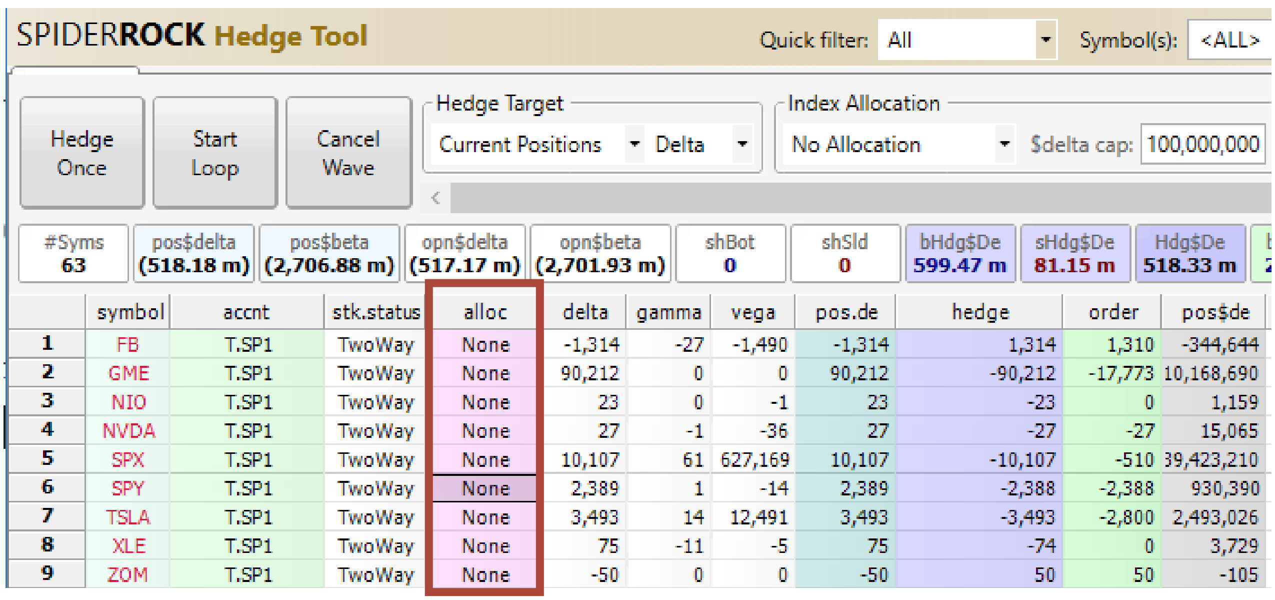
  Important note: The Hedge Tool should be used carefully by users who autohedge any options orders using the Symbol Viewer.

- **Index Allocation**

  Users have a choice to hedge a position naturally with the corresponding underlyer instrument (no allocation as displayed in the dropdown list below), or via a proxy instrument (Comp to Target: Uniform/Comp to Target: Day Gm).



  Below is an example of a portfolio that is hedged with the corresponding underlyer instruments. Please note that the default value under the alloc column is "None".



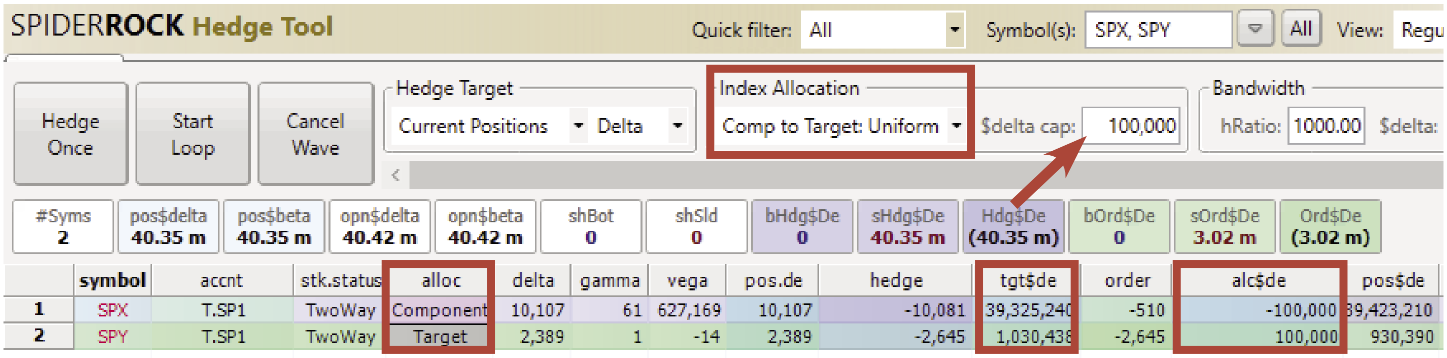
	symbol	accnt	stk.status	alloc	delta	gamma	vega	pos.de	hedge	order	pos$de
1	FB	T.SP1	TwoWay	None	-1,314	-27	-1,490	-1,314	1,314	1,310	-344,644
2	GME	T.SP1	TwoWay	None	90,212	0	0	90,212	-90,212	-17,773	10,168,690
3	NIO	T.SP1	TwoWay	None	23	0	-1	23	-23	0	1,159
4	NVDA	T.SP1	TwoWay	None	27	-1	-36	27	-27	-27	15,065
5	SPX	T.SP1	TwoWay	None	10,107	61	627,169	10,107	-10,107	-510	39,423,210
6	SPY	T.SP1	TwoWay	None	2,389	1	-14	2,389	-2,388	-2,388	930,390
7	TSLA	T.SP1	TwoWay	None	3,493	14	12,491	3,493	-3,493	-2,800	2,493,026
8	XLE	T.SP1	TwoWay	None	75	-11	-5	75	-74	0	3,729
9	ZOM	T.SP1	TwoWay	None	-50	0	0	-50	50	50	-105

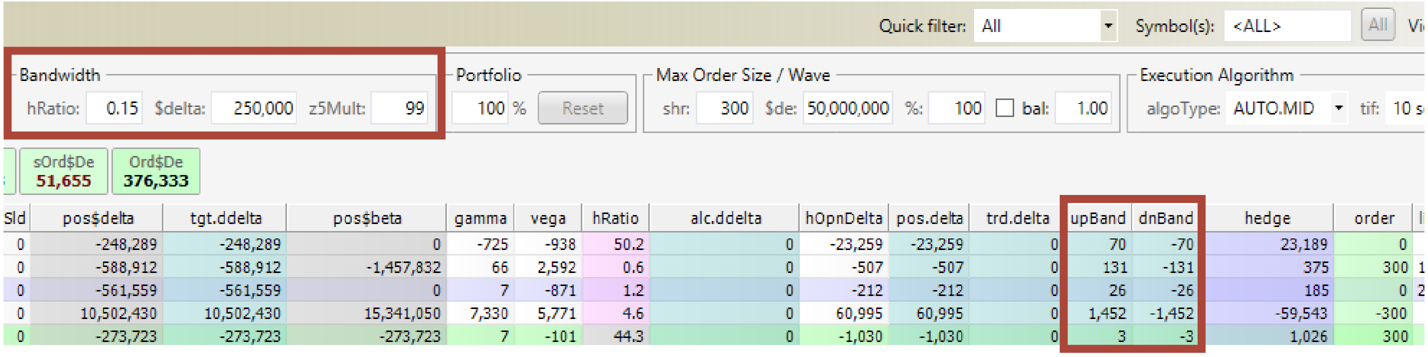
  If users wanted to allocate the deltas of a symbol position and hedge these deltas with a different symbol, they can select either 'Comp to Target: Uniform' or 'Comp to Target: Day Gm' to proxy hedge.

  - 'Comp to Target: Uniform' will allocate the maximum of the symbol delta position or the $ delta cap amount to the target symbol uniformly.
  - 'Comp to Target: Day Gm' will allocate the maximum of the symbol delta position or the $ delta cap amount to the target symbol gamma-weighted.

To set the Target symbol and Component symbol(s), users right-click on the 'alloc' cell of a specific symbol to select the appropriate value.

Below is an example where the user has chosen an Index Allocation of 'Component to Target: Uniform' setting SPX to **Component** and SPY to **Target.** By setting a $delta cap of $100,000 (see red arrow), the user is allocating a maximum of 100,000 $deltas from SPX to SPY to hedge the SPX position.





- **Bandwidth**

  The bandwidth values (hRatio, $delta, z5Mult) enable users to define a risk tolerance. These values will create an up and down delta band to limit the hedging down to the nearest band. The number of shares needed to be bought/sold to hedge to the nearest band are displayed in the hedge column. Bandwidth values can be combined, but the tightest value will prevail to determine the up and down band.

- **hRatio**

  Hedge Ratio (hRatio) is a value that represents the average 1-day move of the underlyer price. ($deltas/$gammas * square root of 252 days/implied vol)

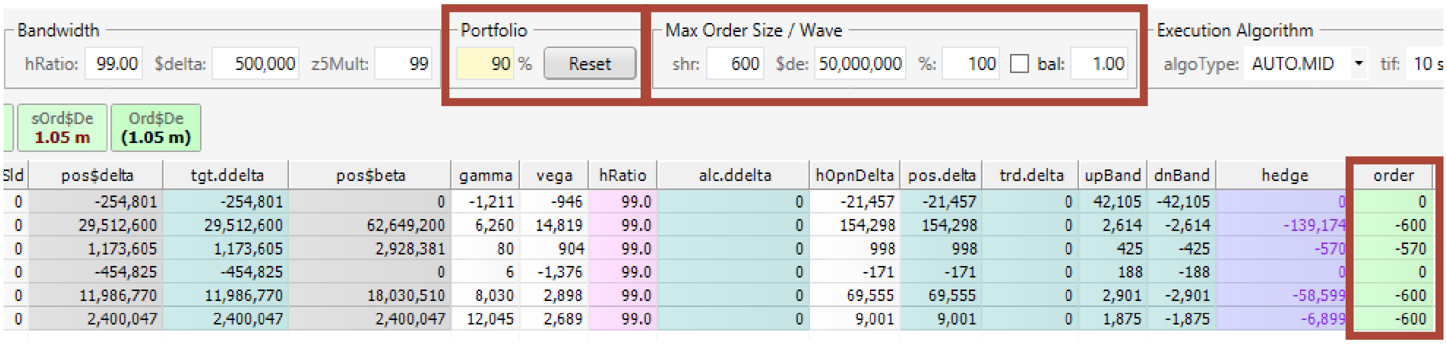
  Example: Assume your position in XYZ is 0 delta and 1000 gamma. If the calculated average daily move in the stock is 1.00 dollar per day, then the position will have swings of 1000 deltas on average. If the hedge ratio is set to "1.0" and the stock moves by 1.00 dollar, you'll have 1000 deltas and the hedge loop will not generate an order as the position is still within the 'bandwidth' of a 1.0 hedge ratio. If stock moves by $1.50, the delta position will increase to 1500 and the hedge loop will generate an order for 500 shares to hedge the XYZ position back to the 1000 delta bandwidth.

- **$delta**

  This sets a risk tolerance measured in dollar deltas (underlyer price * delta).

- **z5Mult**

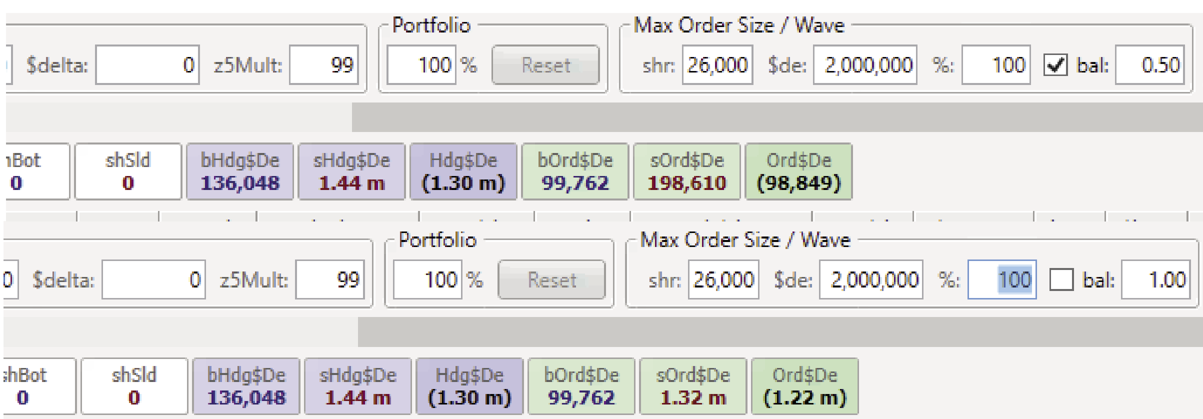
  The Z5 bandwidth is a mathematical attempt to optimize the trade-off between the cost of hedging and the error in replicating a risk-free payoff (as in Black-Scholes original paper with ‘continuous’ rebalancing to replicate the payoff of a treasury). When a position’s actual hedge exceeds a bandwidth around it’s theoretical net delta, the position is re-hedged to the nearest bandwidth, not the theoretical delta (de).



- **Portfolio**

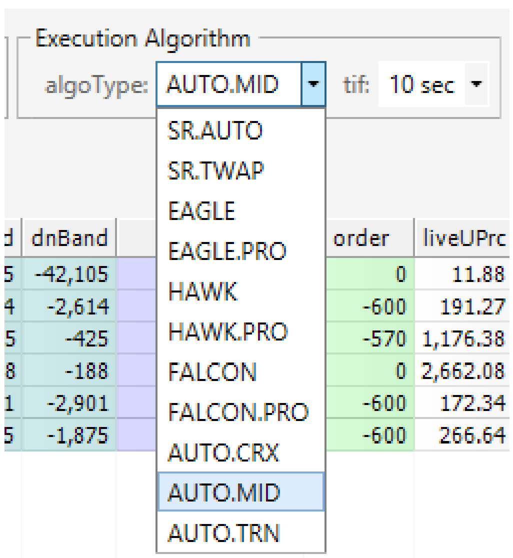
  This box allows the user to set what percentage of their portfolio positions they would like to hedge.

- **Max Order Size / Wave**
  - shr: The shares box (shr) allows the user to set a maximum number of shares per order for each wave that the system is sending when hedging. In the example above, the user has set a maximum of 600 shares per order which is reflected in the order column. When the hedge target is below the number of shares, the system will create an order with a size corresponding to the hedge target.
  - $de: The $delta box ($de) allows users to set a maximum $delta per wave sent to the system.
  - %: This is the percentage of the portfolio to be hedged.
  - bal: CChecking this box will balance the portfolio to be hedged based on a buy-to-sell ratio.
    - A bal ratio of 0.5 will balance twice as many sells as buys.
    - A bal ratio of 1 will balance buys and sells equally.
    - A balance ratio of 2 will balance twice as many buys as sells.



The purple boxes above display the current position $delta (buy, sell and net) of the portfolio. The values in the green boxes show the anticipated $delta impact before hedging. Modifying the bal ratio would change the values in those green boxes significantly as the hedging would lean differently based on this ratio.

- **Execution Algorithm**
  - AlgoType: A variety of stock algos are available for users to work their orders more or less aggressively.
  - SR.AUTO: Will take and make liquidity cycling from a passive to aggressive mode
  - SR.TWAP: Same as above but will work in a TWAP fashion
  - EAGLE, HAWK and FALCON will take liquidity when conditions are optimal based on the SpiderRock alpha probabilities. Eagle being the most passive and Falcon being the most aggressive.
  - EAGLE.PRO, HAWK.PRO and FALCON.PRO work as above, but can post liquidity based on alpha probabilities in addition to taking liquidity.
  - AUTO.CRX: Limit price goes out on the bid price if selling/on the ask price if buying
  - AUTO.MID: Limit price goes out at mid-market and is restated every 5 seconds
  - AUTO.TRN: Limit price turns the market by one price increment and is restated every 20 seconds



  - Tif: The time-in-force dropdown allows a user to select the amount of time a basket of orders will work, either once or in waves depending on whether the user has selected Hedge Once or Start Loop. The time frame ranges from 10 seconds to 30 minutes or extended day.

- **Quick filter**

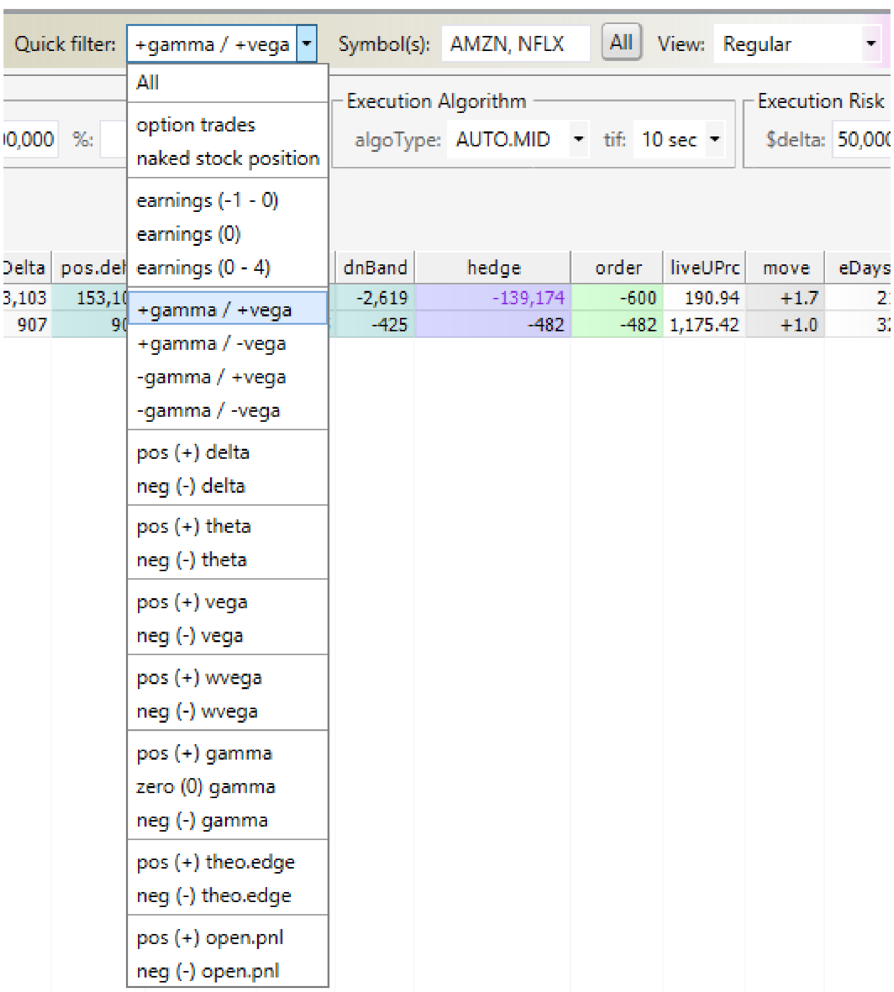
  Users can apply filters to their portfolio to isolate the positions with certain risk characteristics. The user can then hedge sections of their portfolio based on different risk tolerance criteria. The list of filters available is shown below. It is possible for users to have several instances of the hedge tool open so that they can create different risk tolerance levels for each sections of the portfolio determined by the filters. For example, symbols with a positive gamma might be hedged differently than symbols with a negative gamma.

- **Symbol(s)**

  Users can also isolate one or several symbols using the Symbol(s) box. Symbols are delimited by a comma as shown below.

- **View**

  The default view is Regular. Users can select Complex to see positions that derive from corporate actions.

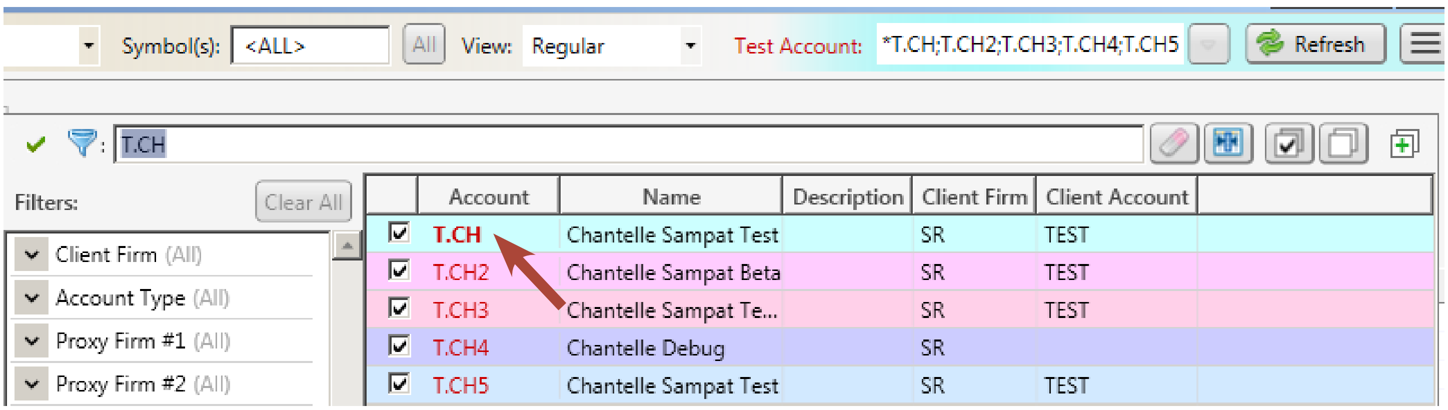


HEDGING USING MULTIPLE ACCOUNTS

Selecting Accounts

Users are able to hedge from multiple accounts into one master account. In order to select the accounts to hedge from, manually type the accounts into the account box followed by a semi colon after each acocunt, or select them from the account selector. Notice in the picture below, that the account 'T.CH' is in bold letters. This is the master account. All hedges will be done from combined positions from all accounts selected and the executions will take place in the master account.

The master account can be changed by double clicking on another account in the account selector or by placing an asterick in front of the desired master account.



The selected master account will be displayed in the accnt column as shown below.